

Scalability & Future Plan

Date	11 July 2026
Team ID	
Project Name	StudyAI-Smart AI educate Companion&Quiz Generator
Maximum Marks	1 Mark

Scalability & Future Plan:

StudyAI currently runs on a lightweight stack Flask backend, Firebase Firestore, and the Groq API for AI generation built to comfortably handle individual and small-group usage. This document looks at what happens as usage grows: where the current free-tier hosting setup would start to strain under heavier traffic, what upgrades would keep response times fast, and which features (like collaborative study groups or a mobile app) are realistic next steps once the core product is validated.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Render free-tier backend spins down when idle	First request after inactivity can take 50+ seconds	High
2	No caching for repeated AI generation requests	Slower response times and higher Groq API usage for similar content	Medium
3	Single-region Firestore deployment	Higher latency for users far from the deployed region	Low

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Handles ~20 concurrent users comfortably (tested via JMeter)	Upgrade to Render's paid tier for always-on instances and higher concurrency
2	Data Storage	Firebase Firestore, suitable for moderate scale	Add indexing and query optimization as document/user count grows
3	Performance	Groq API calls add most of the latency	Cache common summaries/flashcards to reduce repeat API calls
4	Security	JWT-based auth, Firebase Authentication	Add rate limiting on auth/upload endpoints to prevent abuse

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 1	Streaks & reminder notifications	1–2 months	Improve daily engagement and retention
Phase 2	Multi-language support for summaries/quizzes	2–3 months	Wider accessibility for non-English speaking students
Phase 3	Collaborative study groups (shared flashcard decks)	3–4 months	Enable peer learning and social study features
Phase 4	Mobile app (React Native)	4–6 months	Reach users who primarily study on mobile